SCHOOLS & COLLEGES · PROGRAMMING, WEB, DATA, HARDWARE

Every course ends with something built.

Learn with purpose. Build with curiosity. Every lesson takes you closer to creating something meaningful. Start small, think big, and keep building the future—one project at a time.



Learn | Build | Solve | Make Impact

PREREQUISITES

The learning pathway

Each course connects to the ones it unlocks. Colour is carried through the rest of the deck.

FOUNDATIONS

C Programming
6 wks · 24 hrs



Python Programming
8 wks · 32 hrs



Web Development
8 wks · 32 hrs



APPLIED

C++ Programming
8 wks · 32 hrs

Java Programming
8 wks · 32 hrs

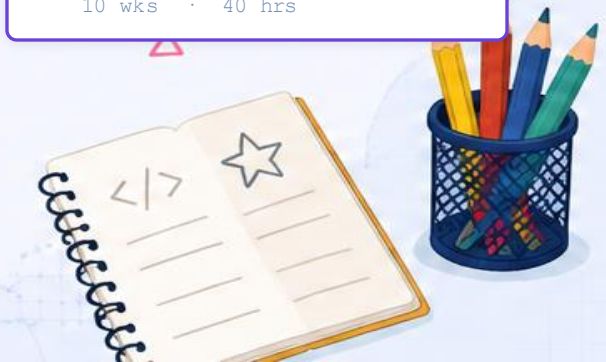
Arduino & Pi
8 wks · 32 hrs

Reactjs
10 wks · 40 hrs

ADVANCED

Data Science
12 wks · 48 hrs

Computer Vision
10 wks · 40 hrs



FOUNDATIONS · BEGINNER

C Programming

4 weeks · 2 hours

CONTENT

01 Data types, variables, operators

02 Control flow: if / else, loops

03 Functions and recursion

04 Arrays, strings and pointers

05 Structures and unions

06 File handling

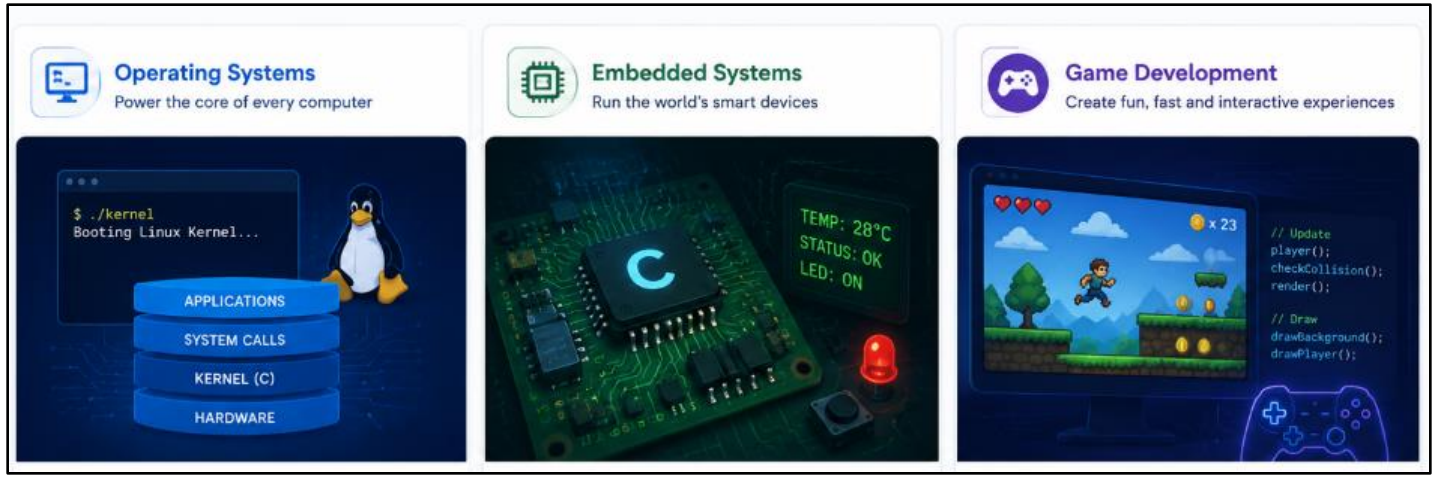
07 Dynamic memory allocation

08 Command Line Arguments

09 Data Structure in C

10 Mini Projects

Real World Systems



Student Record System

- **Operating Systems:** C is used to develop kernels, device drivers, memory management, file systems, and other core system components.
- **Embedded Systems:** C powers microcontrollers and smart devices such as appliances, robots, IoT devices, automotive systems, and medical equipment.
- **Game Development:** C provides the performance and control required for game engines, graphics, physics, audio, and real-time gameplay.

APPLIED · BEGINNER-INTERMEDIATE

C++ Programming

8 weeks · 2 hours

CONTENT

01 C++ fundamentals, syntax, references

02 Classes, objects

03 Encapsulation, Constructors, destructors

04 Inheritance and polymorphism

05 Operator and function overloading

06 Exception handling and file streams

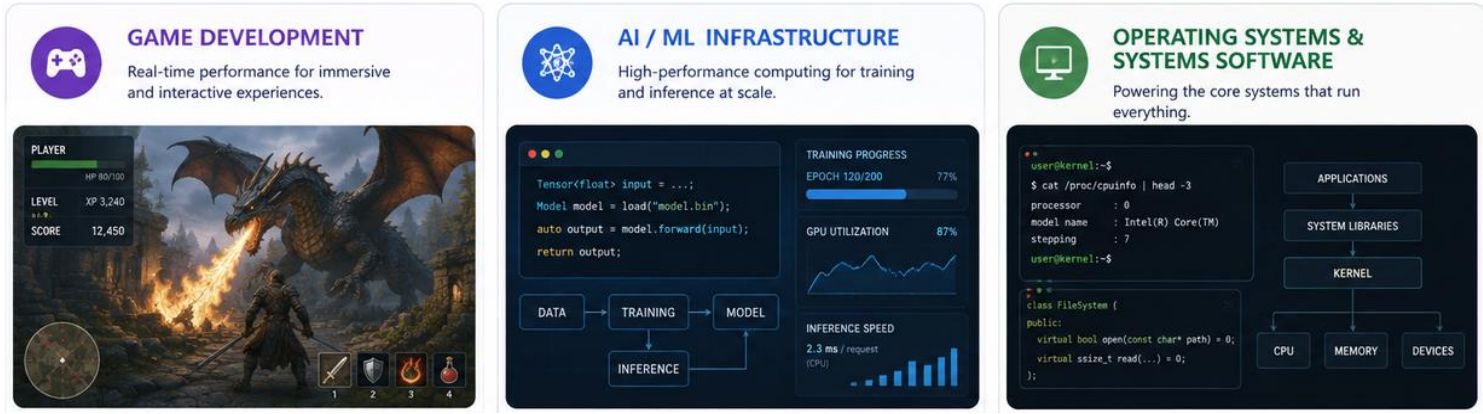
07 Smart pointers and memory management

08 Templates and the STL

09 Algorithms, iterators and lambda functions

10 Mini projects and application design

REAL WORLD PROGRAMS



Operating Systems & Systems Software: C++ is used to build system components such as kernels, device drivers, compilers, networking systems, file systems, and performance-critical software.

AI / ML Infrastructure: C++ provides the high-performance computing required for machine-learning frameworks, inference engines, tensor operations, GPU acceleration, and large-scale model deployment.

Game Development: C++ powers game engines and provides the performance and control required for graphics, physics, audio, animation, and real-time gameplay.

FOUNDATIONS · BEGINNER-ADVANCED

Python Programming

10 weeks · 2 hours

CONTENT

01 Python syntax, variables and data types

02 Operators, conditions and control flow

03 Functions, scope and lambda functions

04 Lists, tuples, dictionaries and sets

05 Comprehensions and string processing

06 Modules, packages and virtual environments

07 File handling and exception handling

08 Object-oriented Python

09 Encapsulation, constructors, destructors

10 Inheritance and polymorphism

11 Operator and function overloading

12 NumPy and Pandas for data processing

13 Web scraping and task scheduling

14 Automation mini-projects

15 Automation major-projects

REAL WORLD PROGRAMS



AI / Machine Learning: Python is used to build AI applications, train machine-learning models, process data, develop computer vision and NLP systems, and deploy intelligent models.

Data Science & Analytics: Python is used to clean, analyze, visualize, and transform data, helping organizations discover patterns, generate insights, and make data-driven decisions.

Automation & Backend: Python is used to automate repetitive tasks, build APIs and web applications, scrape and process data, and create backend services and scheduled workflows.

APPLIED · BEGINNER-INTERMEDIATE

Java Programming

8 weeks · 32 hours

CONTENT

01 Java fundamentals, syntax and data types

02 Control flow, loops and arrays

03 Classes, objects and encapsulation

04 Inheritance, interfaces and polymorphism

05 Collections Framework: List, Set and Map

06 Exception handling and file I/O

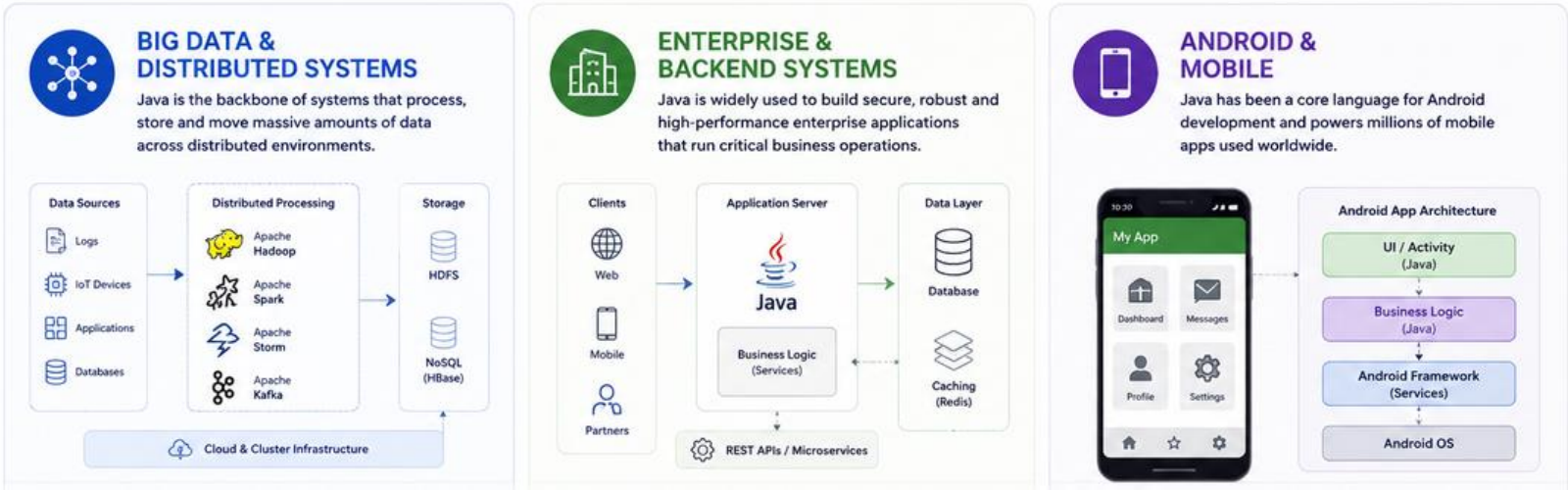
07 Generics, enums and packages

08 Multithreading and concurrency basics

09 JDBC and database connectivity

10 Desktop GUI and application projects

REAL WORLD PROGRAMS



Big Data & Distributed Systems: Java is used to build systems that process, store, and move massive amounts of data, including distributed processing platforms, data pipelines, real-time streaming, and large-scale storage systems.

Enterprise & Backend Systems: Java powers reliable, scalable backend applications such as banking systems, payment platforms, REST APIs, business applications, microservices, and enterprise software.

Android & Mobile: Java has been widely used to develop Android applications, providing the foundation for mobile apps, business applications, utilities, games, and other services running on Android devices.

FOUNDATIONS · BEGINNER-INTERMEDIATE

Web Development

8 weeks · 2 hours

CONTENT

01 HTML basics and page structure

02 HTML headings, paragraphs, links and images

03 CSS basics, colors, fonts and spacing

04 CSS Flexbox and basic layouts

05 Responsive web design

06 JavaScript basics, variables and data types

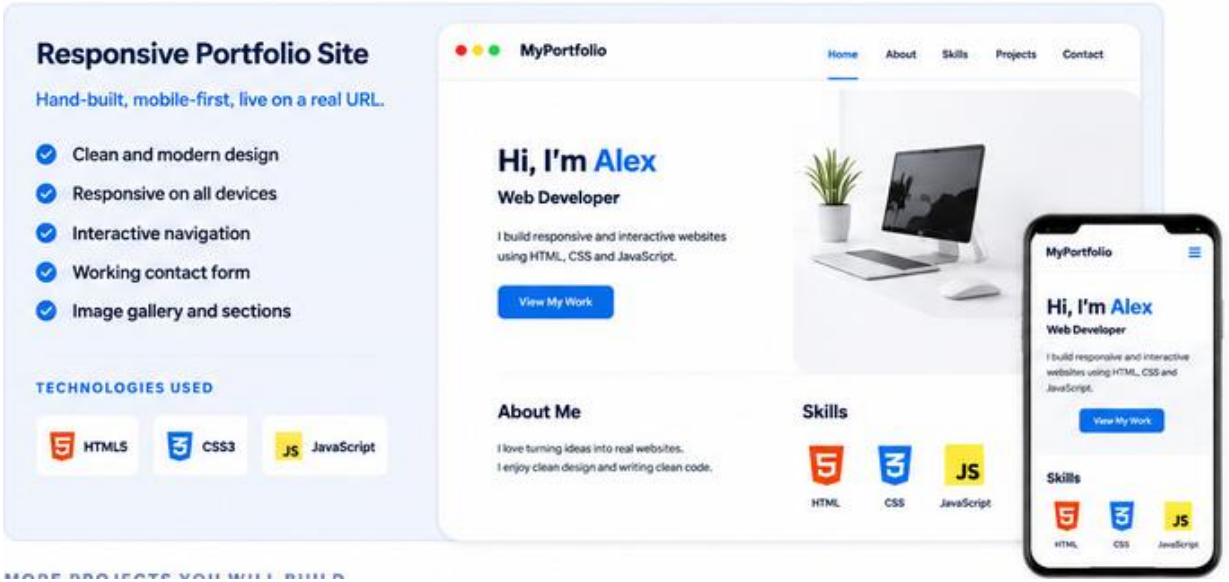
07 Conditions, loops and functions

08 DOM manipulation and events

09 Forms and basic validation

10 Fetch API, JSON and mini projects

OUTCOME — WHAT THEY WALK OUT WITH



Responsive Portfolio Site: A responsive personal portfolio website built from scratch using HTML, CSS, and JavaScript, featuring a modern layout, responsive navigation, project sections, skills showcase, image gallery, and an interactive contact form.

School Event Landing Page: A responsive event website with structured sections, event details, registration form, and mobile-friendly layouts.

Weather App: A web application that fetches real-time weather data from a public API and displays location, temperature, conditions, and other weather information.

APPLIED· BEGINNER-INTERMEDIATE

React-JS

8 weeks · 2 hours

CONTENT

01 React fundamentals and project setup

02 JSX, components and props

03 State and event handling

04 Conditional rendering and lists

05 Forms and user input

06 React Hooks: `useState` and `useEffect`

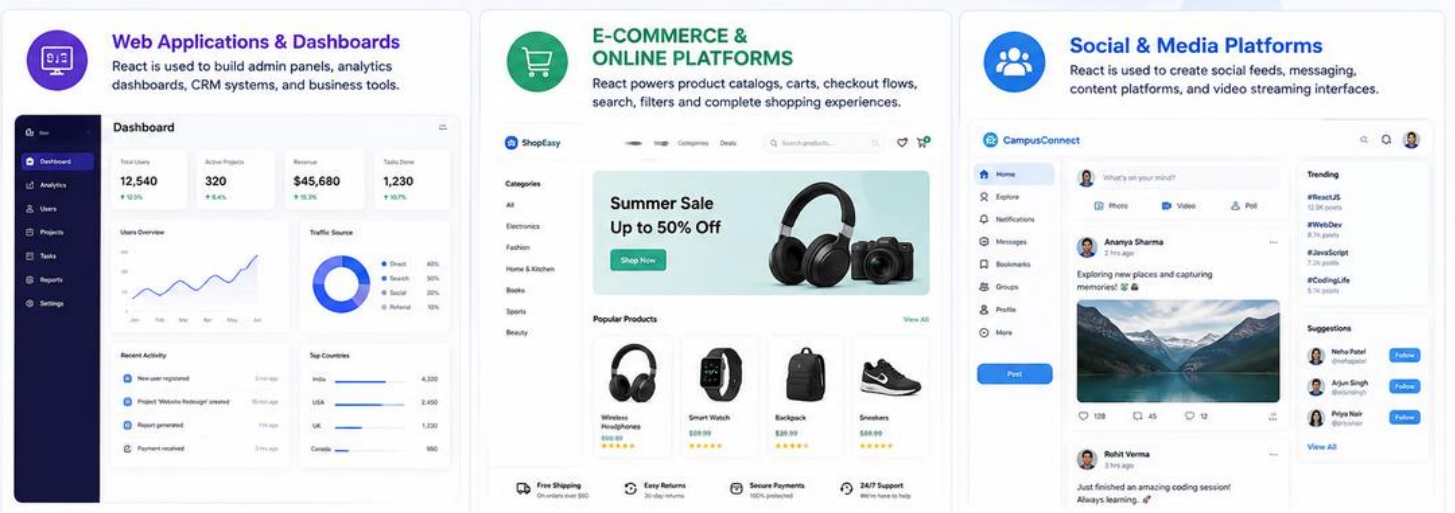
07 Fetching APIs and working with JSON

08 Component reuse and basic routing

09 Building and organizing a React application

10 Final React projects

REAL WORLD PROGRAMS



Web Applications & Dashboards: ReactJS is used to build interactive web applications such as admin dashboards, analytics platforms, student portals, CRM systems, project-management tools, and business applications.

E-Commerce & Online Platforms: ReactJS powers dynamic shopping experiences including product catalogs, search and filtering, shopping carts, checkout interfaces, customer accounts, and online marketplaces.

Social & Media Platforms: ReactJS is used to create responsive interfaces for social feeds, messaging platforms, content-sharing applications, video platforms, and online communities.

APPLIED · BEGINNER-INTERMEDIATE

Arduino & Raspberry Pi

8 weeks · 32 hours

CONTENT

01 Introduction to IOT

02 Board anatomy, GPIO pins and basic wiring

03 Digital input and output

04 Analog input and output

05 LEDs, buttons and buzzers

06 Basic sensors: temperature, light and distance

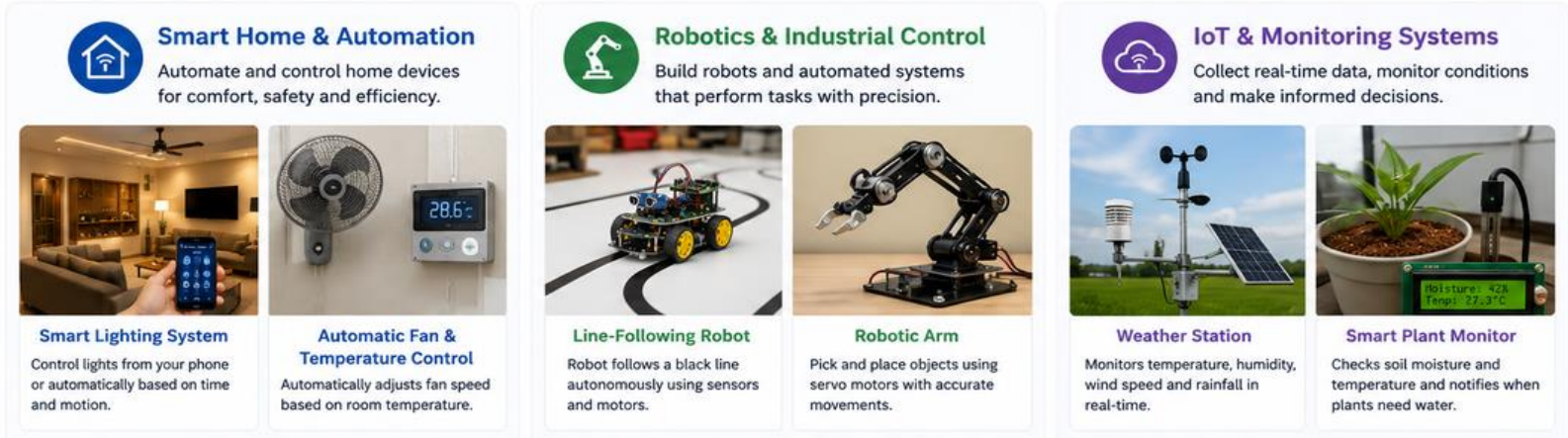
07 Motors, servos and relays

08 Arduino programming and serial monitor

09 Raspberry Pi with Python and GPIO

10 Mini projects and final build

REAL WORLD PROGRAMS



Smart Home & Automation: Arduino and Raspberry Pi are used to automate everyday devices such as smart lights, fans, door locks, security alarms, and temperature-control systems.

Robotics & Industrial Control: Arduino and Raspberry Pi are used to build robots and automated machines that control motors, servos, sensors, robotic arms, and conveyor systems.

IoT & Monitoring Systems: Arduino and Raspberry Pi are used to collect and monitor real-time sensor data for applications such as weather stations, smart plant systems, air-quality monitors, water-quality monitoring, and environmental tracking.

ADVANCED · INTERMEDIATE-ADVANCED

Data Science

14 weeks · 2 hours

CONTENT

01 Python for Data Science fundamentals

02 NumPy arrays and numerical computing

03 Pandas for data manipulation

04 Data loading, exploration and profiling

05 Data cleaning and preprocessing

06 Data wrangling, merging and grouping

07 Data visualization with Matplotlib and Seaborn

08 Descriptive statistics and probability

09 Exploratory Data Analysis (EDA)

10 Feature engineering and data preparation

11 Machine learning with Scikit-learn

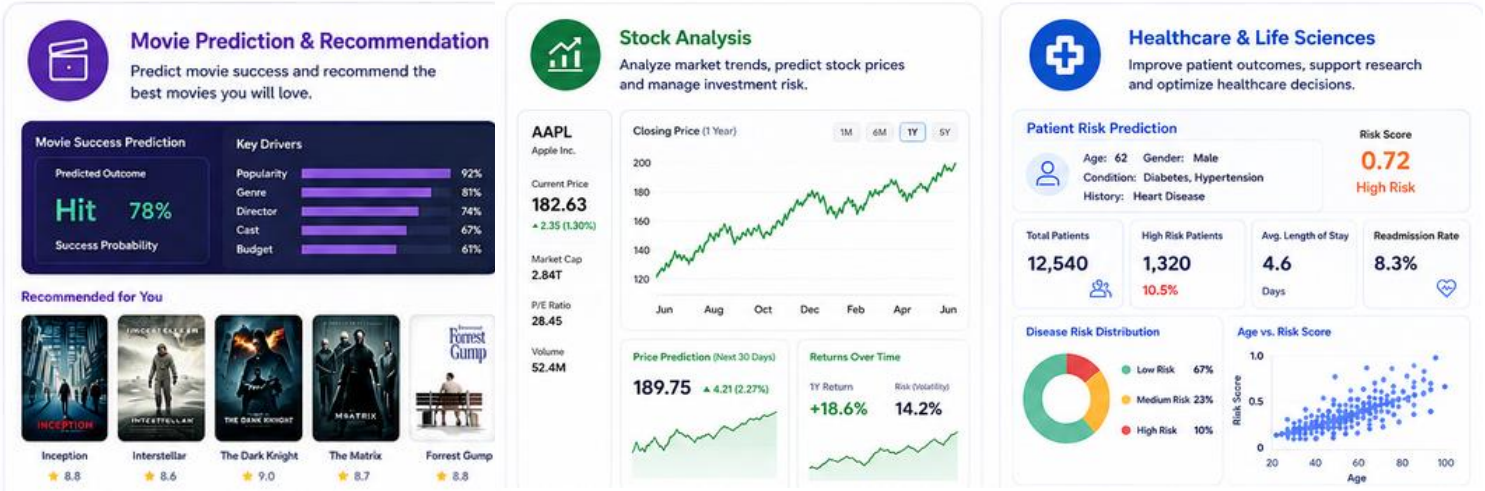
12 Regression and prediction

13 Classification and clustering

14 Model evaluation, validation and tuning

15 End-to-end Data Science projects

REAL WORLD PROGRAMS



Movie Prediction & Recommendation: Data science is used to analyze movie ratings, genres, user preferences, and viewing behavior to predict ratings, identify audience interests, and recommend movies that users are likely to enjoy.

Stock Analysis: Data science is used to analyze historical prices, trading volumes, market trends, and financial indicators to identify patterns, evaluate risk, and support data-driven investment analysis.

Healthcare & Life Sciences: Data science is used to analyze patient and clinical data to identify health patterns, predict risks, support medical research, improve diagnosis, and help healthcare organizations make better decisions.

ADVANCED · ADVANCED

Computer Vision

14 weeks · 2 hours

CONTENT

01 Introduction to Computer Vision and images

02 OpenCV basics: reading, displaying and saving images

03 Image properties, pixels, colors and channels

04 mage resizing, cropping and transformations

05 Image filtering, blurring and edge detection

06 Thresholding, contours and shape detection

07 Object and face detection with OpenCV

08 Feature extraction and image matching

09 Video processing and webcam applications

10 Object tracking and motion detection

11 Introduction to machine learning for images

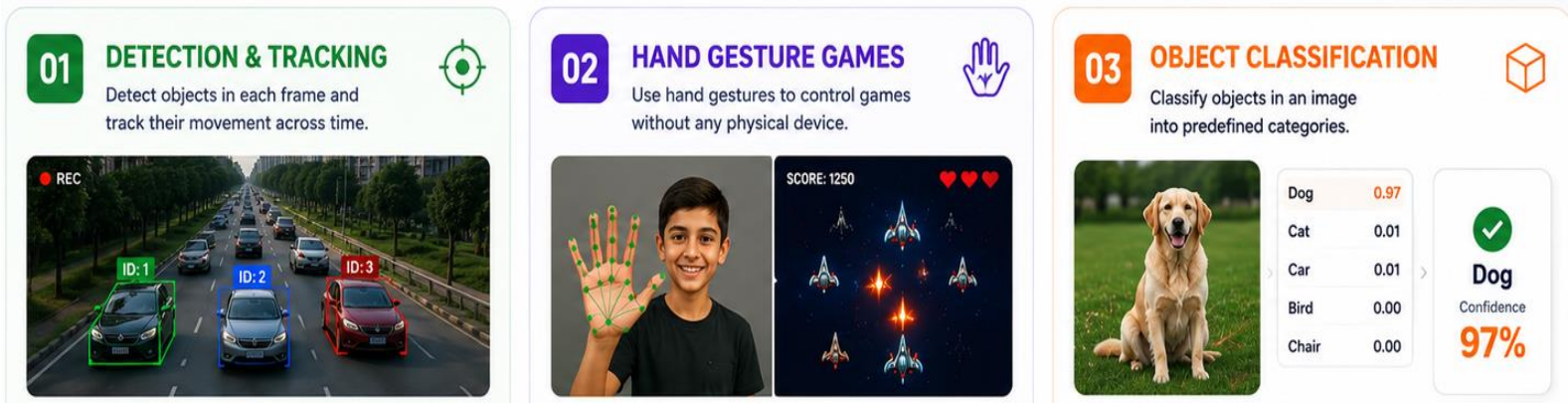
12 Convolutional Neural Networks (CNNs)

13 Image classification with deep learning

14 Real-time object detection

15 End-to-end Computer Vision projects

REAL WORLD PROGRAMS



Detection & Tracking: Computer vision is used to detect and track people, vehicles, and other objects in real time for applications such as traffic monitoring, surveillance, sports analytics, and autonomous systems.

Hand Gesture Games: Computer vision enables cameras to recognize hand movements and gestures, allowing users to control games and interactive applications without physical controllers.

Object Classification: Computer vision models classify objects in images and video using techniques such as image processing and convolutional neural networks, supporting applications such as image search, retail analytics, medical imaging, and wildlife monitoring.

SECTION 03

Projects that get used

Each build solves a problem a school, college, home or local shop actually has — so the project keeps running after the course ends, and it holds up in an interview or a science fair.

PRJ-01	**Face-Recognition Attendance**	School office
PRJ-02	**Smart Classroom Power Saver**	Campus facilities
PRJ-03	**Digital Notice Board**	College admin
PRJ-04	**Exam Result Analyser**	Teachers
PRJ-05	**Water Tank Level Monitor**	Hostel / home
PRJ-06	**Shop Billing & Stock System**	Local shop
PRJ-07	**Library Book Tracker**	School library
PRJ-08	**Plant Disease Detector**	Agriculture / fair
PRJ-09	**Air Quality Monitor**	Campus / fair
PRJ-10	**School Bus Tracker**	Parents

SECTION 03 · 1 OF 3

Real-world builds

Course tags show where the skills come from.

PRJ-01

School office

Face-Recognition Attendance

Roll-call eats ten minutes of every period and the register is still copied by hand.

- Pi camera marks the register as students walk in
- Writes a timestamped sheet the office can open
- Flags unrecognised faces for manual check

Python Programming | Computer Vision | Arduino & Pi

PRJ-02

Campus facilities

Smart Classroom Power Saver

Fans and lights run in empty rooms all afternoon and nobody notices the bill.

- Motion sensor cuts the relay after a set idle time
- Logs unit-hours saved each week
- Manual override switch for exam halls

C Programming | Arduino & Pi

PRJ-03

College admin

Digital Notice Board

Paper notices get missed, torn down, or never reach the students who needed them.

- Staff post notices from a Django admin panel
- Students see a live feed by department
- Push alerts for urgent items

Web Development | React & Django

PRJ-04

Teachers

Exam Result Analyser

Marks sit in spreadsheets and nobody has time to work out which topics the class failed.

- Reads and cleans the marks sheet automatically
- Charts weak subjects per class and student
- Flags students slipping across tests

Python Programming | Data Science

Real-world builds

Course tags show where the skills come from.

PRJ-05

Hostel / home

Water Tank Level Monitor

The overhead tank runs dry without warning, or overflows and wastes a full tank.

- Ultrasonic sensor reads the level continuously
- Buzzer and phone alert at low and full marks
- Optional auto cut-off for the motor

Arduino & Pi | Python Programming

PRJ-06

Local shop

Shop Billing & Stock System

A neighbourhood store bills on paper and finds out about stock-outs from customers.

- Fast billing screen with running total
- Stock deducts on every sale, alerts when low
- Daily and monthly sales summary

C++ Programming | Java Programming

PRJ-07

School library

Library Book Tracker

Issue registers are hard to search and overdue books are chased weeks late.

- Issue and return by ISBN or barcode scan
- Automatic overdue list every morning
- Search by title, author or student

Java Programming | React & Django

PRJ-08

Agriculture / fair

Plant Disease Detector

Farmers and gardeners spot leaf disease late, after it has already spread.

- Phone photo of a leaf goes to a trained CNN
- Returns likely disease and confidence score
- Keeps a log of what was found and when

Python Programming | Data Science | Computer Vision

Real-world builds

Course tags show where the skills come from.

PRJ-09

Campus / fair

Air Quality Monitor

Nobody knows whether the air in the workshop or near the road is actually safe.

- Pi logs dust, temperature and humidity all day
- Live web chart anyone can open on a phone
- Exports the history for a report

Arduino & Pi | Python Programming | Data Science | Web Development

PRJ-10

Parents

School Bus Tracker

Parents wait at the stop with no idea whether the bus is five minutes or forty away.

- GPS module on the Pi reports position
- Parents watch the bus on a live map
- Alert when the bus is near their stop

Arduino & Pi | Python Programming | React & Django

Why these projects work

1 **They span courses**

Most builds need two or three courses, so the pathway pays off instead of feeling like separate subjects.

2 **They stay in use**

The school, hostel or shop keeps running the project after the batch ends.

3 **They demo well**

A working device or live dashboard holds up in an interview, a science fair or a college application.

NEXT STEPS

Pick a track, pick a batch length.

Every duration here is a starting point. Tell me the batch length and the year group and the sequence can be rebuilt around it.

Semester

14-16 weeks, one course per group, weekly lab

Summer bootcamp

4-6 weeks intensive, two courses back to back

Weekend batch

One course over 3-4 months, Saturday sessions

9 courses · 312 hours · 9 capstones · 10 real-world builds